

Student Name: Pragati Agrawal

Roll Number: 220779

Date: April 16, 2026

1. **Error in Summary:** "The architecture consists of an Encoder network that maps the input $\mathbf{x}$ directly to a single point vector $\mathbf{z}$ in a lower-dimensional latent space."

Correction: Instead of mapping an input to a static vector, the encoder (recognition model) generates the parameters for a probability distribution. The latent variable $\mathbf{z}$ is then sampled from this distribution to represent the input.

Evidence from Paper: [Section 2.1] "In this paper we will therefore also refer to the recognition model $q_\phi(\mathbf{z}|\mathbf{x})$ as a probabilistic encoder, since given a datapoint $\mathbf{x}$ it produces a distribution (e.g. a Gaussian) over the possible values of the code $\mathbf{z}$ from which the datapoint $\mathbf{x}$ could have been generated."

2. **Error in Summary:** "the authors introduce the Reparameterization Trick, which involves adding uniform noise (sampled from $U[-1,1]$) to the latent vector before it is passed to the decoder."

Correction: The trick works by expressing the latent variable as a differentiable function of the input and an auxiliary noise variable (ϵ) - $\mathbf{z} = g_\phi(\epsilon, \mathbf{x})$. For the standard VAE, this noise is drawn from a standard Normal distribution $\mathcal{N}(0, 1)$ rather than a uniform distribution.

Evidence from Paper: [Section 2.4] "The essential parameterization trick is quite simple. Let z be a continuous random variable, and $\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})$ be some conditional distribution. It is then often possible to express the random variable $\mathbf{z}$ as a deterministic variable $\mathbf{z} = g_\phi(\epsilon, \mathbf{x})$," "In this case, a valid reparameterization is $\mathbf{z} = \mu + \sigma\epsilon$ where ϵ is an auxiliary noise variable $\epsilon \sim \mathcal{N}(0, 1)$."

3. **Error in Summary:** "and a KL Divergence term that acts as a penalty to ensure the approximate posterior $q_\phi(\mathbf{z}|\mathbf{x})$ is as dissimilar as possible from the prior $p(\mathbf{z})$."

Correction: The KL divergence term in the ELBO objective functions as a regularizer that minimizes the difference between the two distributions. Its goal is to make the approximate posterior align closely with the prior, not to distance them.

Evidence from Paper: [Section 2.3] "The KL-divergence term can then be interpreted as regularizing ϕ , encouraging the approximate posterior to be close to the prior $p_\theta(\mathbf{z})$."

4. **Error in Summary:** "Finally, the paper demonstrates through experiments on the Frey Faces dataset that for complex image data, the prior $p(\mathbf{z})$ must be a learned mixture of Gaussians to achieve high-quality reconstruction."

Correction: The researchers actually utilized a fixed, non-learned centered isotropic Gaussian as the prior for their experiments on both the MNIST and Frey Faces datasets.

Evidence from Paper: [Section 5] "... gradients are computed by differentiating the lower bound estimator $\nabla_{\theta, \phi} \mathcal{L}(\theta, \phi; \mathbf{X})$ (see algorithm 1), plus a small weight decay term corresponding to a prior $p(\theta) = \mathcal{N}(0, \mathbf{I})$ "

Student Name: Pragati Agrawal

Roll Number: 220779

Date: April 16, 2026

1. **Error in Summary:** "The authors propose an algorithm that only requires the user to provide the analytic gradient of the ELBO with respect to the latent variables z ."
Correction: The BBVI approach removes the need for manual analytic derivations; practitioners simply need to provide a function for the model's log-likelihood, and the framework estimates gradients using Monte Carlo samples drawn from the variational distribution.
Evidence from Paper: [Section 1] "From the practitioner's perspective, this method requires only that he or she write functions to evaluate the model log-likelihood. The remaining calculations (sampling from the variational distribution and evaluating the Monte Carlo estimate) are easily put into a library to be shared across models..."
2. **Error in Summary:** "In this paper, control variates are used solely to bias the gradient estimates toward the global optimum, ensuring that the stochastic updates never stray into local minima."
Correction: Control variates are implemented to minimize the variance of the gradient estimator rather than to introduce bias. Because the technique preserves the original expectation, the estimator remains unbiased and does not offer a mechanism to bypass local minima.
Evidence from Paper: [Section 3.2] "As we saw above, variance reduction methods work by replacing the function whose expectation is being approximated by Monte Carlo with another function that has the same expectation but smaller variance. That is, to estimate $E_q[f]$ via Monte Carlo we compute the empirical average of $\hat{f}$ where $\hat{f}$ is chosen so $E_q[f] = E_q[\hat{f}]$ and $Var_q[f] > Var_q[\hat{f}]$."
3. **Error in Summary:** "These functions must be manually derived for every new model to ensure the gradient variance is exactly zero."
Correction: In this framework, control variates rely solely on the score function of the variational distribution, making them model-independent and eliminating the need for manual derivation for each new application. Additionally, while these methods lower variance, the paper does not suggest they result in zero variance.
Evidence from Paper: [Section 1, 3.1] "The remaining calculations ... are easily put into a library to be shared across models, which means our method can be quickly applied to new modeling settings." "... We exploit the structure of our problem to use these methods in a way that requires no model-specific derivations"
4. **Error in Summary:** "To optimize the parameters, the algorithm uses a fixed, large learning rate."
Correction: Instead of a static rate, the algorithm uses the AdaGrad method to provide adaptive learning rates. This approach dynamically adjusts step sizes for individual components based on the historical sum of the outer products of the gradients.
Evidence from Paper: [Section 4] "Intuitively, we would like the learning rate to be small when the variance of the gradient is large and vice-versa. Additionally, in problems

like ours that have different scales, the learning rate needs to be set small enough to handle the smallest scale. To address this issue, we use the AdaGrad algorithm (Duchi et al., 2011). Let G_t be a matrix containing the sum across the first t iterations of the outer products of the gradient. AdaGrad defines a per component learning rate as $\rho_t = \eta \text{diag}(G_t)^{-1/2}$

Student Name: Pragati Agrawal

Roll Number: 220779

Date: April 16, 2026

1. **Error in Summary:** "The authors propose a modification to standard SGD where a **fixed** amount of constant variance noise is added to the gradient updates at every iteration."

Correction: Rather than using a static noise level, SGLD utilizes Gaussian noise where the variance is tied to the learning rate ϵ_t . This step size is **systematically reduced** (annealed) as the training progresses, ensuring the noise level adapts accordingly over time.

Evidence from Paper: [Section 3] "... use Robbins-Monro stochastic gradients, add an amount of Gaussian noise balanced with the step size used, and allow step sizes to go to zero. The proposed update is simply:

$$\Delta\theta_t = \frac{\epsilon_t}{2} \left(\nabla \log p(\theta_t) + \frac{N}{n} \sum_{i=1}^n \nabla \log p(x_{t_i} | \theta_t) \right) + \eta_t$$

$$\eta_t \sim N(0, \epsilon_t)$$

where the step sizes decrease towards zero ..."

2. **Error in Summary:** "... the algorithm eliminates the need for the Metropolis-Hastings (MH) acceptance step by ensuring the step size ϵ_t stays constant throughout training. This allows the chain to maintain a high rejection rate to avoid local minima."

Correction: The rejection step in Metropolis-Hastings becomes unnecessary because the learning rate ϵ_t is gradually reduced to zero. This decay ensures that discretization errors and the likelihood of rejection decrease asymptotically. Furthermore, the goal is not to skip local minima via high rejection, but to sample the area surrounding them.

Evidence from Paper: [Section 3] "... as $\epsilon_t \rightarrow 0$, the discretization error of Langevin dynamics will be negligible so that the MH rejection probability will approach 0 and we may simply ignore this step."

3. **Error in Summary:** "The authors prove that as long as the noise is sampled from a Uniform distribution, the distribution of the parameters will converge to a single MAP (Maximum A Posteriori) estimate."

Correction: The noise is actually drawn from a Gaussian distribution. Consequently, the method is designed to provide samples from the complete posterior distribution rather than converging to a single MAP point estimate.

Evidence from Paper: [Section 1] "Our method combines Robbins-Monro type algorithms ... with Langevin dynamics which injects noise into the parameter updates in such a way that the trajectory of the parameters will converge to the full posterior distribution rather than just the maximum a posteriori ..."

"Langevin dynamics ... injects Gaussian noise into the parameter updates so that they do not collapse to just the MAP solution ..."

4. **Error in Summary:** "... the paper demonstrates the method's efficiency using a Logistic Regression model on the MNIST dataset."

Correction: Although a Logistic Regression model is indeed used to test the algorithm's efficiency, the experiments were actually conducted on the a9a dataset (extracted from the UCI adult dataset) instead of the MNIST dataset.

Evidence from Paper: [Section 5.2] "We applied our inference algorithm to the a9a dataset derived by (Lin et al., 2008) from the UCI adult dataset."

Student Name: Pragati Agrawal

Roll Number: 220779

Date: April 16, 2026

- Error in Summary:** "... deep neural networks (DNNs) with a finite number of hidden units per layer are equivalent to Gaussian Processes."

Correction: The equivalence between deep neural networks and Gaussian Processes is a mathematical limit that is only reached when the width of the hidden layers tends toward infinity.

Evidence from Paper: [Section 1] "As we will discuss in detail below, in the limit of infinite width the Central Limit Theorem implies that the function computed by the neural network (NN) is a function drawn from a Gaussian process (GP). In the case of single hidden-layer networks, the form of the kernel of this GP is well known."
- Error in Summary:** "The paper introduces a recursive method to compute the Neural Tangent Kernel (NTK), which defines the covariance between inputs in a deep GP."

Correction: The recursive method developed by the authors is specifically designed to compute the covariance function for the Neural Network GP (NNGP), not the Neural Tangent Kernel (NTK).

Evidence from Paper: [Section 1.2] "We begin by specifying the form of a GP which corresponds to a deep, infinitely wide neural network— hereafter referred to as the Neural Network GP (NNGP)— in terms of a recursive, deterministic computation of the kernel function. "
- Error in Summary:** "They prove that to maintain this GP equivalence, one must use a Sigmoid activation function, as ReLU activations cause the covariance to collapse to zero at every layer. "

Correction: The NNGP equivalence is a general framework applicable to generic point-wise nonlinearities and is not limited to Sigmoid functions. Contrary to the claim of covariance collapse, the authors explicitly demonstrate that ReLU activations result in a well-defined covariance function known as the arccosine kernel.

Evidence from Paper: [Section 2.3] "In the case of the ReLU non linearity, it yields the well-known arccosine kernel (Cho & Saul (2009)) whose form we reproduce in Appendix B. When no analytic form exists, it can instead be efficiently computed numerically, as described in Section 2.5."
- Error in Summary:** "Additionally, the authors find that the GP performance is always strictly worse than the corresponding finite-width trained neural network. Finally, the researchers evaluate their theory using the MNIST dataset, showing that the GP derived from a 10-layer CNN outperforms a standard SVM"

Correction: Contrary to the summary, the researchers found that NNGP performance is consistently competitive with and often exceeds the performance of finite-width neural networks trained with gradient-based methods.

Evidence from Paper: [Section 3.1]

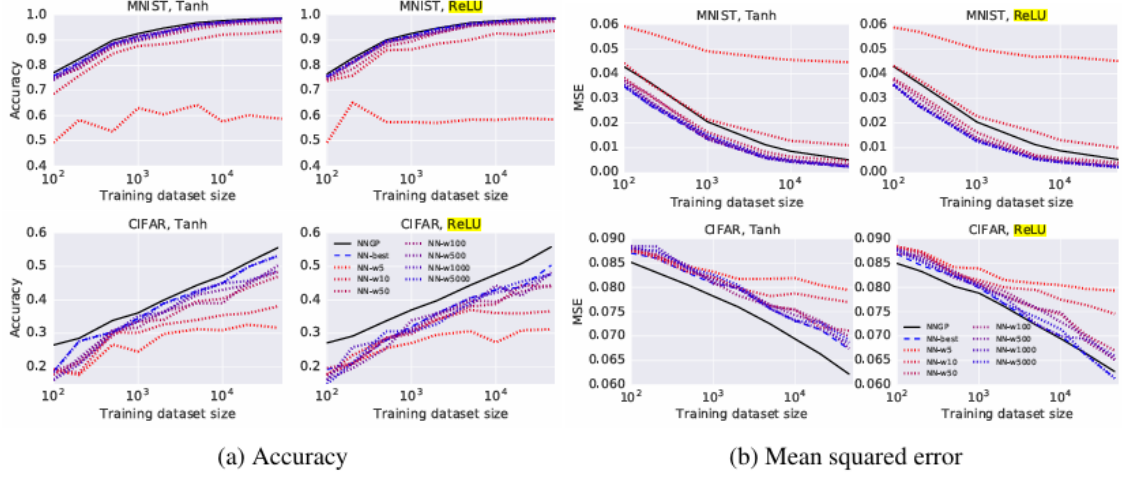


Figure 1: The NNGP often outperforms finite width networks, and neural network performance more closely resembles NNGP performance with increasing width. Test accuracy and mean squared error on MNIST and CIFAR-10 dataset are shown for the best performing NNGP and best performing SGD trained neural networks for given width. ‘NN-best’ denotes the best performing (on the validation set) neural network across all widths and trials. Often this is the neural network with the largest width.

Table 1: The NNGP often outperforms finite width networks. Test accuracy on MNIST and CIFAR-10 datasets. The reported NNGP results correspond to the best performing depth, σ_w^2 , and σ_b^2 values on the validation set. The traditional NN results correspond to the best performing depth, width and optimization hyperparameters. Best models for a given training set size are specified by (depth-width- σ_w^2 - σ_b^2) for NNs and (depth- σ_w^2 - σ_b^2) for GPs. More results are in Appendix Table 2.

Num training	Model (ReLU)	Test accuracy	Model (tanh)	Test accuracy
MNIST:1k	NN-2-5000-3.19-0.00	0.9252	NN-2-1000-0.60-0.00	0.9254
	GP-20-1.45-0.28	0.9279	GP-20-1.96-0.62	0.9266
MNIST:10k	NN-2-2000-0.42-0.16	0.9771	NN-2-2000-2.41-1.84	0.9745
	GP-7-0.61-0.07	0.9765	GP-2-1.62-0.28	0.9773
MNIST:50k	NN-2-2000-0.60-0.44	0.9864	NN-2-5000-0.28-0.34	0.9857
	GP-1-0.10-0.48	0.9875	GP-1-1.28-0.00	0.9879
CIFAR:1k	NN-5-500-1.29-0.28	0.3225	NN-1-200-1.45-0.12	0.3378
	GP-7-1.28-0.00	0.3608	GP-50-2.97-0.97	0.3702
CIFAR:10k	NN-5-2000-1.60-1.07	0.4545	NN-1-500-1.48-1.59	0.4429
	GP-5-2.97-0.28	0.4780	GP-7-3.48-2.00	0.4766
CIFAR:45k	NN-3-5000-0.53-0.01	0.5313	NN-2-2000-1.05-2.08	0.5034
	GP-3-3.31-1.86	0.5566	GP-3-3.48-1.52	0.5558

Student Name: Pragati Agrawal

Roll Number: 220779

Date: April 16, 2026

1. **Error in Summary:** "To solve this, BatchBALD maximizes the sum of the individual mutual information of all points in a candidate batch. "

Correction: BatchBALD does not simply sum the individual mutual information scores of candidate points, as that approach fails to account for redundancy within the batch. Instead, the algorithm maximizes the joint mutual information between the entire batch of points and the model parameters.

Evidence from Paper: [Section 3.1] We propose BatchBALD as an extension of BALD whereby we jointly score points by estimating the mutual information between a joint of multiple data points and the model parameters.

$$a_{\text{BatchBALD}}(\{\mathbf{x}_1, \dots, \mathbf{x}_b\}, p(\boldsymbol{\omega} \mid \mathcal{D}_{\text{train}})) = \mathbb{I}(y_1, \dots, y_b; \boldsymbol{\omega} \mid \mathbf{x}_1, \dots, \mathbf{x}_b, \mathcal{D}_{\text{train}}).$$

2. **Error in Summary:** This objective ensures that the points in the batch are as similar to each other as possible to provide a 'consensus' update to the model parameters θ .

Correction: The BatchBALD objective is designed to select a diverse set of points rather than similar ones. By jointly scoring the points in a batch, the algorithm ensures that the selected batch provides a broad cover of new information.

Evidence from Paper: [Section 3.2]

BatchBALD, on the other hand, takes overlaps into account by computing $\mu^*(\bigcup_i y_i \cap \boldsymbol{\omega})$ and is more likely to acquire a more diverse cover under maximisation:

$$\mathbb{I}(y_1, \dots, y_b; \boldsymbol{\omega} \mid \mathbf{x}_1, \dots, \mathbf{x}_b, \mathcal{D}_{\text{train}}) = \mathbb{H}(y_{1:b} \mid \mathbf{x}_{1:b}, \mathcal{D}_{\text{train}}) - \mathbb{E}_{p(\boldsymbol{\omega} \mid \mathcal{D}_{\text{train}})} \mathbb{H}(y_{1:b} \mid \mathbf{x}_{1:b}, \boldsymbol{\omega}, \mathcal{D}_{\text{train}}) \quad (6)$$

$$= \mu^*\left(\bigcup_i y_i\right) - \mu^*\left(\bigcup_i y_i \setminus \boldsymbol{\omega}\right) = \mu^*\left(\bigcup_i y_i \cap \boldsymbol{\omega}\right) \quad (7)$$

3. **Error in Summary:** "The paper proves that this objective is linearly scalable, meaning that calculating a batch of size 1000 is no more computationally expensive than calculating a batch of size 1"

Correction: BatchBALD is not linearly scalable in a way that makes large batches cost the same as single-point acquisitions; its complexity grows with the batch size b .

Evidence from Paper: [Section 3.4] Monte-Carlo sampling bounds the time complexity of the full BatchBALD algorithm to $\mathcal{O}(bc \cdot \min\{c^b, m\} \cdot |\mathcal{D}_{\text{pool}}| \cdot k)$ compared to $\mathcal{O}(c^b \cdot |\mathcal{D}_{\text{pool}}|^b \cdot k)$ for naively finding the exact optimal batch and $\mathcal{O}((b+k) \cdot |\mathcal{D}_{\text{pool}}|)$ for BALD.

4. **Error in Summary:** Furthermore, the authors demonstrate that BatchBALD is particularly effective for Support Vector Machines (SVMs) because they lack the 'overconfidence' found in Neural Networks.

Correction: The paper is focused on active learning for Deep Bayesian Neural Networks, not SVMs (no mention of SVMs). The authors argue that previous batch acquisition methods fail because they select redundant points, a problem that BatchBALD solves by jointly scoring points for diversity.

Evidence from Paper: [Section 2.3] In this paper we focus on BNNs as our Bayesian model because they scale well to high dimensional inputs, such as images. Compared to regular neural networks, BNNs maintain a distribution over their weights instead of point estimates. Performing exact inference in BNNs is intractable for any reasonably sized model, so we resort to using a variational approximation.

- Error in Summary:** "This paper provides a theoretical justification for deep ensembles by proving that they are equivalent to exact Bayesian inference in neural networks. "

Correction: Deep ensembles are not proven to be equivalent to exact Bayesian inference. They are presented as an empirical alternative that often outperforms Bayesian Neural Networks (BNNs). Furthermore, the authors suggest that deep ensembles' success is better explained by their ability to explore different modes of the loss landscape.

Evidence from Paper: [Abstract] "While deep ensembles were theoretically motivated by the bootstrap, non-bootstrap ensembles trained with just random initialization also perform well in practice, which suggests that there could be other explanations for why deep ensembles work well. Bayesian neural networks, which learn distributions over the parameters of the network, are theoretically well-motivated by Bayesian principles, but do not perform as well as deep ensembles in practice, particularly under dataset shift. "
- Error in Summary:** "... shows that models obtained along a single SGD training trajectory explore multiple distinct modes of the posterior. In contrast, models initialized independently tend to converge to the same mode due to the strong convexity of the loss function."

Correction: The paper's analysis of the loss landscape reveals that models trained along a single SGD trajectory remain within the same low-loss region and produce very similar functions, rather than exploring multiple modes. Conversely, independent random initializations are required to reach distinct, distant modes in the non-convex loss landscape, which is the actual source of functional diversity in deep ensembles.

Evidence from Paper: [Section 4.1] "Figure 2(c) shows that the functions explored by different trajectories (denoted by circles with different colors) are far away, while functions explored within a single trajectory (circles with the same color) tend to be much more similar."
- Error in Summary:** "The authors introduce a diversity-accuracy tradeoff and conclude that increasing diversity always harms predictive accuracy, making large ensembles undesirable."

Correction: The authors demonstrate that diversity is a key driver of ensemble performance. They show that ensembling and subspace sampling provide complementary benefits, and that the advantages of ensembling actually increase when predictions are averaged over more diverse solutions rather than similar ones.

Evidence from Paper: [Section 5] "Figure 8 shows the results on CIFAR-10. The results validate our hypothesis that (i) subspace sampling and ensembling provide complementary benefits, and (ii) the relative benefits of ensembling are higher as it averages predictions over more diverse solutions."
- Error in Summary:** "Finally, the experiments demonstrate that subspace sampling methods such as SWAG consistently outperform deep ensembles on out-of-distribution generalization tasks."

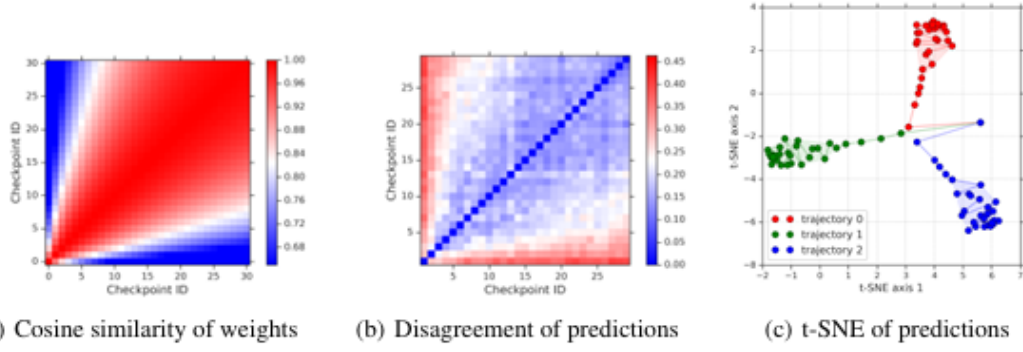


Figure 2: Results using SmallCNN on CIFAR-10. *Left plot*: Cosine similarity between checkpoints to measure weight space alignment along optimization trajectory. *Middle plot*: The fraction of labels on which the predictions from different checkpoints disagree. *Right plot*: t-SNE plot of predictions from checkpoints corresponding to 3 different randomly initialized trajectories (in different colors).

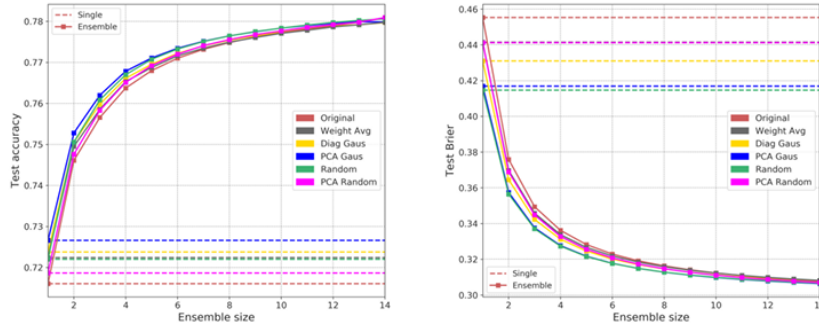


Figure 8: Results using MediumCNN on CIFAR-10 showing the complementary benefits of ensemble and subspace methods as a function of ensemble size. We used 10 samples for each subspace method.

Correction: Experiments actually show that deep ensembles perform better than sampling methods on out-of-distribution tasks and under dataset shift due to higher functional density.

Evidence from Paper: [Section 5] ”Right subplot of Figure 9 shows the results on CIFAR-10-C for increasing corruption intensity. We observe that Jensen-Shannon divergence is the highest between independent random initializations, and lower for subspace sampling methods; the difference is higher under dataset shift, which explains the findings of Ovadia et al. [2019] that deep ensembles outperform other methods under dataset shift.

”

Student Name: Pragati Agrawal

Roll Number: 220779

Date: April 16, 2026

1. **Error in Summary:** "... expressing the Evidence Lower Bound (ELBO) in a transformed parameter space where all latent variables follow a standard normal prior."

Correction: ADVI does not assume or require that latent variables follow a standard normal prior in the transformed space. It uses the change-of-variables formula to transform the original model's joint density $p(x, \theta)$ into a new joint density $p(x, \zeta)$ in an unconstrained real-valued space.

Evidence from Paper: [Section 2.3] "ADVI transforms the model into one with unconstrained real-valued latent variables. Specifically, it transforms $p(x, \theta)$ to $p(x, \zeta)$, where the mapping from θ to ζ is built into the joint distribution. This removes all original constraints on the latent variables θ ."

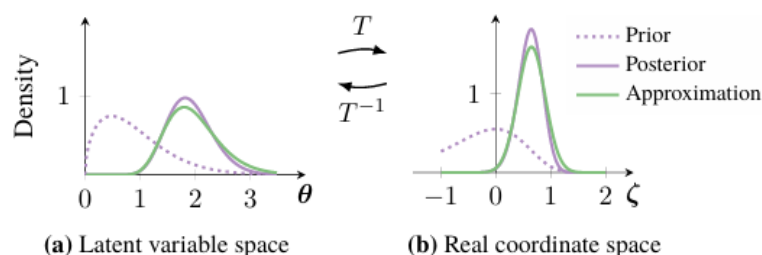


Figure 1: Transforming the latent variable to real coordinate space. The purple line is the posterior. The green line is the approximation. (a) The latent variable space is $\mathbb{R}_{>0}$. (a**→**b) T transforms the latent variable space to $\mathbb{R}$. (b) The variational approximation is a Gaussian in real coordinate space.

2. **Error in Summary:** The method relies on a reparameterization of the variational distribution that eliminates the need for stochastic gradient estimates, allowing exact gradients of the ELBO to be computed via automatic differentiation.

Correction: ADVI is fundamentally a stochastic algorithm that relies on the "reparameterization trick" to approximate the gradient of the ELBO using Monte Carlo samples, rather than computing exact gradients.

Evidence from Paper: [Section 2.2] "We compute the elbo for any model using Monte Carlo (mc) integration, which only requires being able to sample from the variational distribution (Section 2.5). Third, we employ stochastic gradient ascent to maximize the elbo and use automatic differentiation to compute gradients without any user input (Section 2.6)."

3. **Error in Summary:** In practice, ADVI is implemented using a mean-field Gaussian approximation, as the paper shows that full-rank covariance structures offer negligible practical benefits.

Correction: The paper introduces and evaluates full-rank ADVI specifically to capture

posterior correlations, and experiments show that full-rank ADVI consistently performs better in accuracy than the mean-field version.

Evidence from Paper: [Section 3.1] "Figure 4 compares the advi methods to the exact posterior. Both procedures correctly identify the mean of the analytic posterior. However, the shape of the mean-field approximation is incorrect. This is because the mean-field approximation ignores off-diagonal terms of the Gaussian covariance."

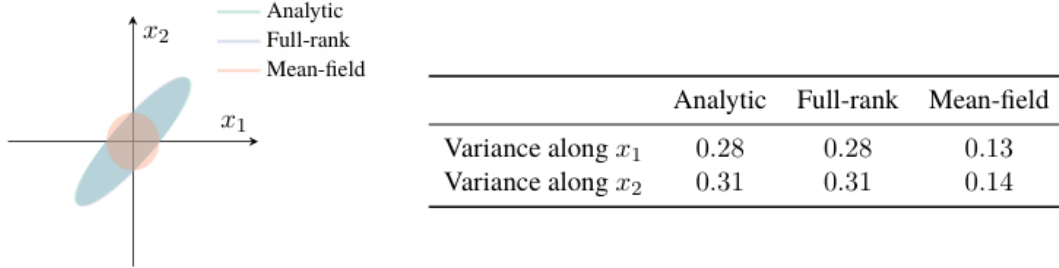


Figure 4: Comparison of mean-field and full-rank ADVI on a two-dimensional Gaussian model. The figure shows the accuracy of the full-rank approximation. Ellipses correspond to two-sigma level sets of the Gaussian. The table quantifies the underestimation of marginal variances by the mean-field approximation.

4. **Error in Summary:** A key advantage of ADVI is that it avoids the need for model-specific derivations by assuming that all latent variables are independent in the variational family, which simplifies inference without significantly affecting posterior correlations.

Correction: The paper introduces and evaluates full-rank ADVI specifically to capture posterior correlations, and experiments show that full-rank ADVI consistently performs better in accuracy than the mean-field version.

Evidence from Paper: [Section 3.1]

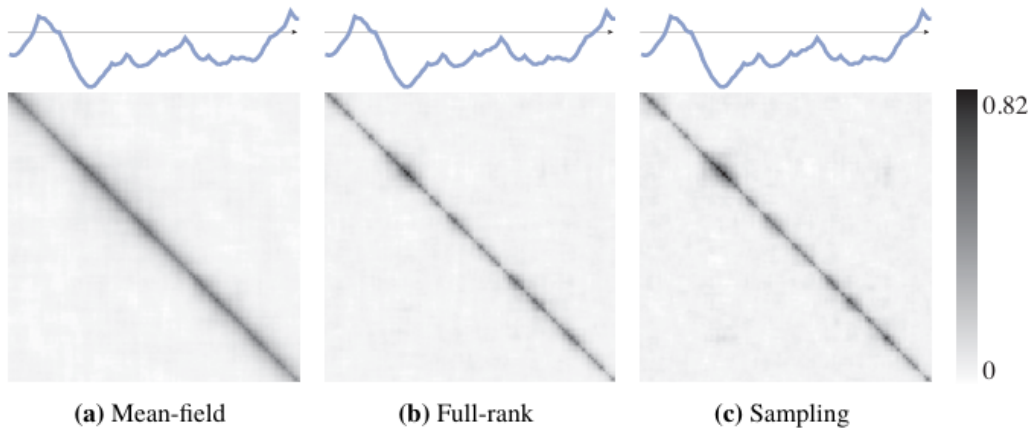


Figure 7: Comparison of empirical posterior covariance matrices. The mean-field ADVI covariance matrix fails to capture the local correlation structure seen in the full-rank ADVI and sampling results. All covariance matrices exhibit a blurry spread due to finite sample size.

1. **Error in Summary:** "In particular, the authors show that different approximations (e.g., diagonal, KFAC, and full-matrix forms) lead to similar predictive performance when applied to modern neural networks."
Correction: The authors demonstrate that the KFAC structured covariance is superior, as it provides a balance between computational efficiency and the expressiveness required for high-quality uncertainty estimation.
Evidence from Paper: [Section 4.1] "The KFAC structured covariance provides a good trade-off between expressiveness and speed. Diagonal approximations perform significantly worse than KFAC and are therefore not suggested. "
2. **Error in Summary:** "A key contribution of the paper is demonstrating that applying LA to only the final layer provides a computationally efficient approximation that closely matches full-network Laplace across a wide range of tasks, including out-of-distribution detection."
Correction: The authors find that applying the Laplace approximation post-hoc to only the last layer typically results in better performance than the full-network version because it suffers less from underfitting. This indicates that last-layer LA is not merely a "matching" approximation but often a superior practical choice, unlike online LA which requires all weights to work reliably.
Evidence from Paper: [Section 4.1] "We find that the online LA only works reliably when it is applied to all weights of the network. In contrast, applying the LA post-hoc only on the last layer instead of all weights typically yields better performance due to less underfitting, and is significantly cheaper. "
3. **Error in Summary:** "The authors further argue that marginal likelihood optimization for tuning prior precision is largely unnecessary in practice, as predictive performance is relatively insensitive to this hyperparameter when using well-trained networks."
Correction: The authors emphasize that the effectiveness of the Laplace approximation is highly dependent on the choice of hyperparameters like prior precision. They demonstrate that optimizing the marginal likelihood (evidence) is a critical and automated way to tune these parameters to ensure the model produces well-calibrated uncertainty estimates, often outperforming the common practice of cross-validation.
Evidence from Paper: [Section 2] "As with all approximate inference methods, the performance of the LA depends on the (hyper)parameters of the prior and likelihood. For instance, it is typically beneficial to tune the prior variance σ^2 used for inference [23, 28, 27, 26, 22]. Commonly, this is done through cross-validation, e.g. by maximizing the validation log-likelihood [23, 48] or, additionally, using out-of-distribution data [28, 49]."
4. **Error in Summary:** "Finally, the paper concludes that Laplace approximations offer competitive uncertainty estimates compared to variational inference and deep ensembles, particularly when computational cost is taken into account, though their performance is largely invariant to the specific predictive approximation used"

Correction: The choice of predictive approximation is not a trivial detail but significantly impacts performance, particularly for out-of-distribution (OOD) detection.

Evidence from Paper: [Section 4.1] ” MC integration can sometimes be superior for OOD detection but at an increased computational cost. The Laplace bridge has the same cost as the probit approximation but typically provides inferior results in our experiments. When using the LA online to optimize hyperparameters, we find that the resulting MAP predictive provides good performance in-distribution, but a probit or MC predictive improves OOD performance.”